

OSSAMA ELHAKKI

Data Scientist / Forecaster — Demand Forecasting · Supply Chain Analytics · ML

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ABOUT ME

Data Scientist with a Master's in Distributed Systems & AI and hands-on expertise in **demand forecasting and supply-chain analytics**. I have benchmarked and tuned 19 forecasting models on 215K+ rows of supply-chain data using **walk-forward cross-validation**, with deep focus on forecast-accuracy measurement (MAE, SMAPE, R^2) and root-cause analysis of variance. Strong in **Python, Pandas, Docker and GitHub**, comfortable in customer-facing settings, and skilled at turning large retail/supply-chain datasets into clear, actionable recommendations.

FEATURED PROJECT — DEMAND FORECASTING

Historical Product Demand Forecasting — DataCo Supply Chain (215K+ rows)

- **19-model benchmark** — classical time-series (Theta, Prophet, ARIMA) → ML (CatBoost, XGBoost, LightGBM, Ridge, Quantile Regression) → deep learning (LSTM, TFT, N-BEATS) → ensembles.
- **Forecast accuracy** — best model CatBoost $R^2=0.7125$; Quantile Regression (P50) best balance (MAE 8.5K, SMAPE 75%, R^2 0.70), far outperforming classical TS baselines.
- **Walk-forward CV & tuning** — expanding-window cross-validation with Optuna hyperparameter search; 36 engineered features; ADF stationarity testing.
- **Root-cause analysis** — diagnosed the ML-vs-classical-TS paradox (correct patterns vs correct scale) and explained why DL underperforms on discrete product-category structure; SHAP for explainability.
- **Reproducible & documented** — published end-to-end on Kaggle with clear methodology and visualizations.

EXPERIENCE

Ofoto — Fine Art Digital Studio

Data Scientist & AI Engineer | Feb 2025 – Oct 2025 | Casablanca

- **Production support & monitoring** — operated and monitored AI services in production with 99.9% uptime, troubleshooting issues and validating outputs daily.
- **Automation with Python** — automated data pipelines and reporting; managed code and deployments with Docker, GitHub and CI/CD.
- **Customer-facing delivery** — translated client needs into solutions, modeled delivery across 5 sprints, and maintained clear documentation.

FSTM Research Laboratory

Data Analyst Intern | Feb 2023 – Jul 2023 | Mohammedia

- Built data-processing pipelines and automated reporting, removing manual work and improving accuracy.

EDUCATION

ENSET Mohammedia — Master's, Distributed Systems & AI

Bien

FST Mohammedia — Bachelor's, Applied Mathematics

Assez Bien

SKILLS

Forecasting	Demand planning, walk-forward CV, forecast accuracy (MAE/SMAPE/ R^2), variance & root-cause analysis
ML / TS Models	CatBoost, XGBoost, LightGBM, Quantile Regression, Prophet/ARIMA/Theta, LSTM, TFT, N-BEATS, Optuna
Programming	Python, Pandas, NumPy, SQL; familiarity with Polars
Dev & Deployment	Docker, GitHub, CI/CD, model monitoring (MLflow, Prometheus/Grafana)
Analysis & Reporting	EDA, SHAP explainability, dashboards (Power BI, Tableau, Plotly), stakeholder reporting
Domain	Supply-chain & retail analytics, replenishment & inventory concepts

LANGUAGES

Arabic — Native · English — B1/B2 · French — B1/B2